Who do you think you could explain the internet to?

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Q1

~Code~
340

CS 340

Network Introduction

Updates

1. MP6 Scheduler Tests due Tuesday

out today

2. HW 7 due Wednesday

3. Final project December 15th at 7:00 - 10:00 pm

MP like specifications

Day of
- participation
- if the code you are running meet specifications

Functions as 1st class objects

a function name is just a reference to a specific function objects

```
x = 4

def func():
    print(5)

x = func
x()
```

Functions as 1st class objects

```
1 def square(x):
2     return x * x
3
4 f = square
5 print(f(4))
```

Functions as 1st class objects

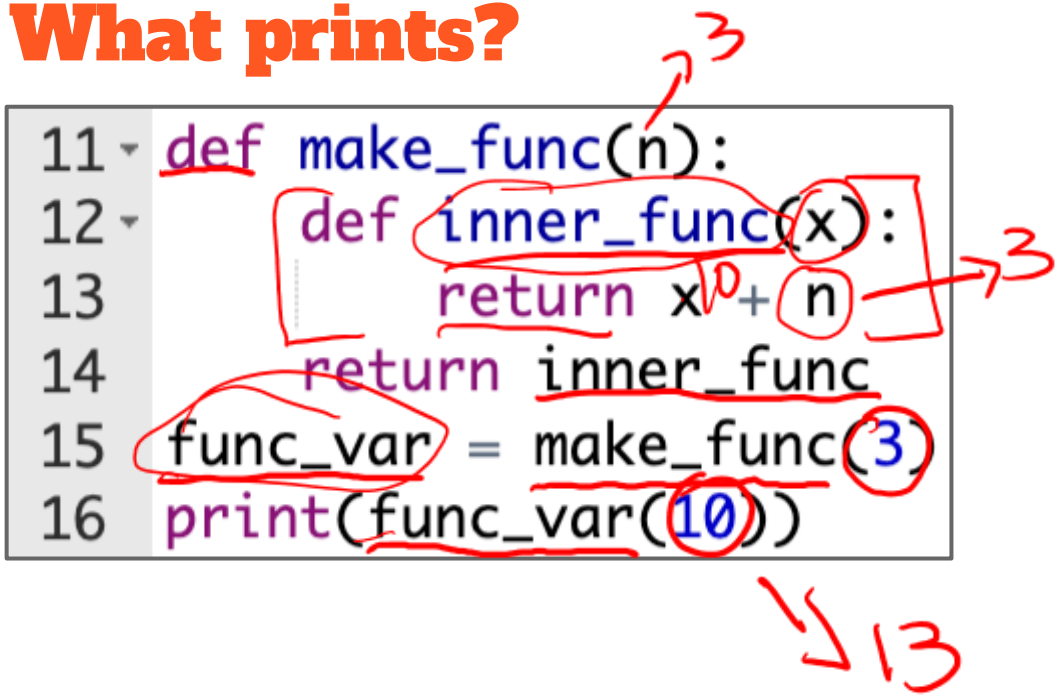
```
1  def square(x):
2      return x * x
```

```
7  def apply(fn, value):
8      return fn(value)
9  print(apply(square, 5))
```

25

What prints?

```
11 def make_func(n):
12     def inner_func(x):
13         return x + n
14     return inner_func
15 func_var = make_func(3)
16 print(func_var(10))
```

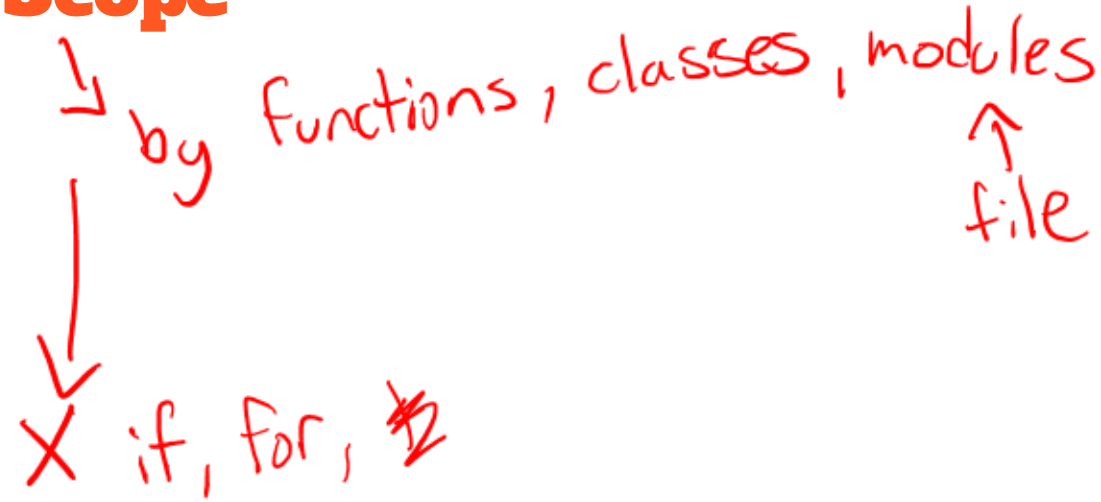


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Python Scope



What happens?

```
2  x = 4
3  def func():
4      global x
5      x += 1
6
7  func()
8  print(x)
```

5

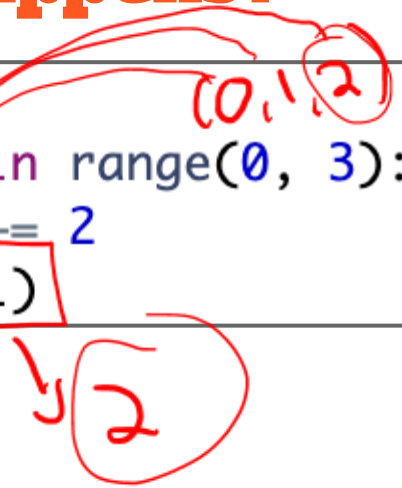
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What happens?

```
2  x = 5
3  for i in range(0, 3):
4      x += 2
5  print(i)
```



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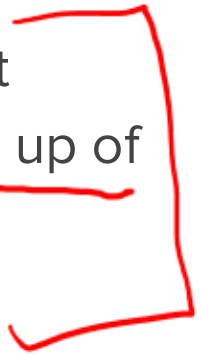
Q4

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Networks

Today's LGs - Fill in the gaps in your understanding of how the internet works

Be able to answer big picture questions about the internet
Be able to answer questions about what a packet is made up of
Be able to differentiate between IP, TCP, and UDP
Be able to explain the client server relationship



Agenda

1. What is a System
2. The internet
    a. IP addresses
    b. Packet format
    c. UPD
    d. TCP

Meet the person next to you!

- What is your name
- A halloween costume you are going to wear or wore at one point
- How would you define the term system?
  - Is a CPU a system?
  - Is a transistor a system?
  - Is a computer a system?

So Far

Individual computer system - parts like CPU RAM, Disk

- VM, thread, scheduling

The OS manages... the computer system.

how processes interact



Internet

The internet is...

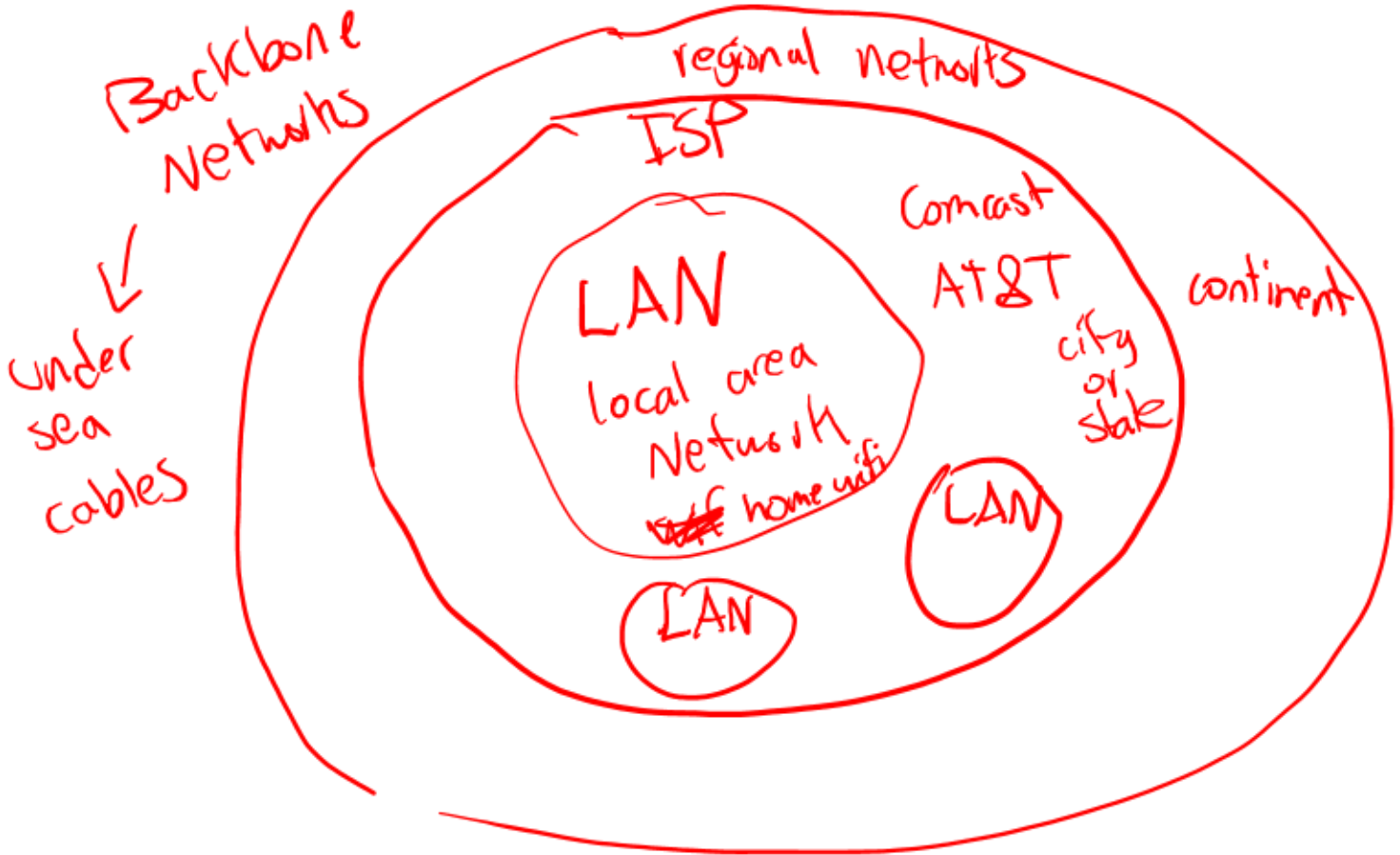
physical infrastructure

connected system of computers and routers

special computer

moves bytes around via wires and radio waves

Network Hierarchy



Internet Protocol (IP)

a unique indentifier for a device on the internet

like an address

IP Addresses

IPv4 - 32 bit addresses 2^{32} ~4.3 billion

IPv6 - 128 bit addresses

IPv4

IPv6

172.16.0.1

2001:0db8:85a3:0000:0000:8a2e:0370:7334

How many IP addresses do you think we need?

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Probably more than 4.3 billion . . .

A) Multiple thousands
B) Multiple million
C) Multiple billion
D) Multiple trillion

If IPv4 with 32 bit addresses can represent ~4 million addresses, how many can IPv6 with a 128 bit address represent?

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Q6

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every grain of sand could have

7.5×10^{18} IP addresses

A) (128/32) * 4 million aka 16 million
B) (128 - 32) * 4 million aka 384 million
C) 2^128 aka 3.4028237e+38
D) 128 trillion

DNS (domain name system)

DNS servers ← computers

you send them a message saying "what is google's IP address"

```
drschatz@cs-drschatz-MBP ~ % nslookup google.com
Server:         2001:558:feed::1
Address:        2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.217.110

drschatz@cs-drschatz-MBP ~ %
```

DNS server IPV6

IPV4

Let's say a router...

Gets the bytes:

0x504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE

- How do we interpret these?

We need a protocol for the format

What do you think is being sent through wires and radio signals from router to router?

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A) 2 signals, either 1 or 0
B) 16 different signals to make up a hex value
C) Voice audio saying a hex value
D) Morse Code

IPv6 Packet

- small chunk of data
  - 40 bytes IP header ← right computer
  - 8, or 20+ bytes TCP/UDP header ← where on your computer
  - 1.4kB (max) - payload
    - HTTP

Talk to your neighbor: Why do you think there is a size limit to packets?

error recovery - if a packet gets dropped or corrupted

fair sharing - prevent something big clogging the network

physical constraints

Talk to your neighbor: Why don't we make packets even smaller?

each packet needs a header

40 bytes + 20 bytes

payload - 2 bytes

Layers of a packet

Layer	Protocols	Use
Application	HTTP, SSH, SMTP, etc.	programmer-visible messages
Transport	**TCP**, **UDP**	reach a specific program
Internet	**IP**, which has two versions: IPv4 and IPv6	reach a specific computer



UDP - User Datagram Protocol

* always need an IP header

+ specify source and destination process
- checksum

Not guaranteed to arrive
or in any order

light weight fast

video calling

TCP - Transmission Control Protocol

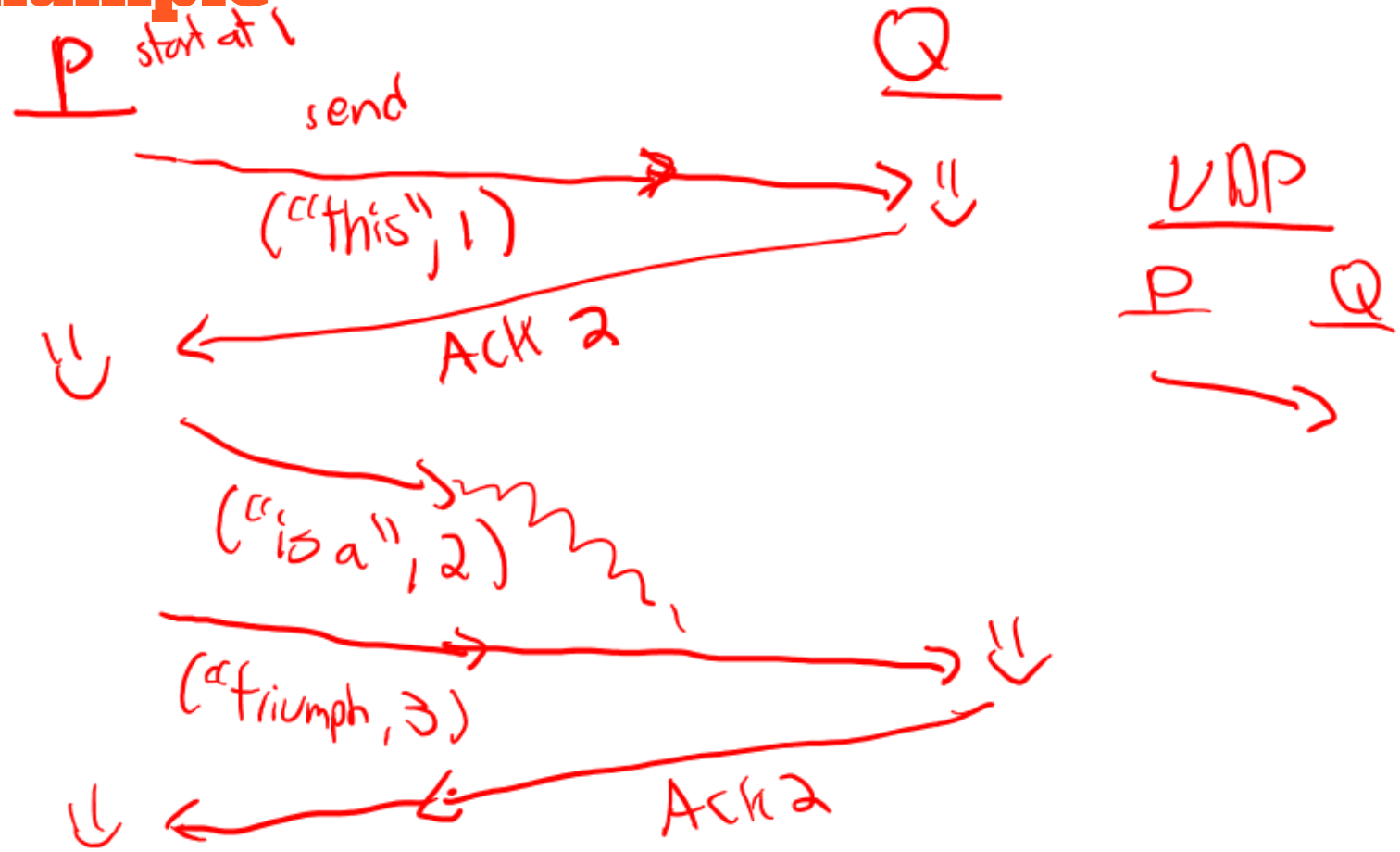
* also with IP layer > also has which process

Reliable and ordered

1st ~~the~~ they have a handshake

- ~~the~~ A to B
  - hello here is my sequence start number
- B to A
  - I see you, here is mine
- A - B
  - I got your sequence number

TCP Example



Which do you think is faster for getting data from a server to your computer?

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a computer

A) IP + UDP

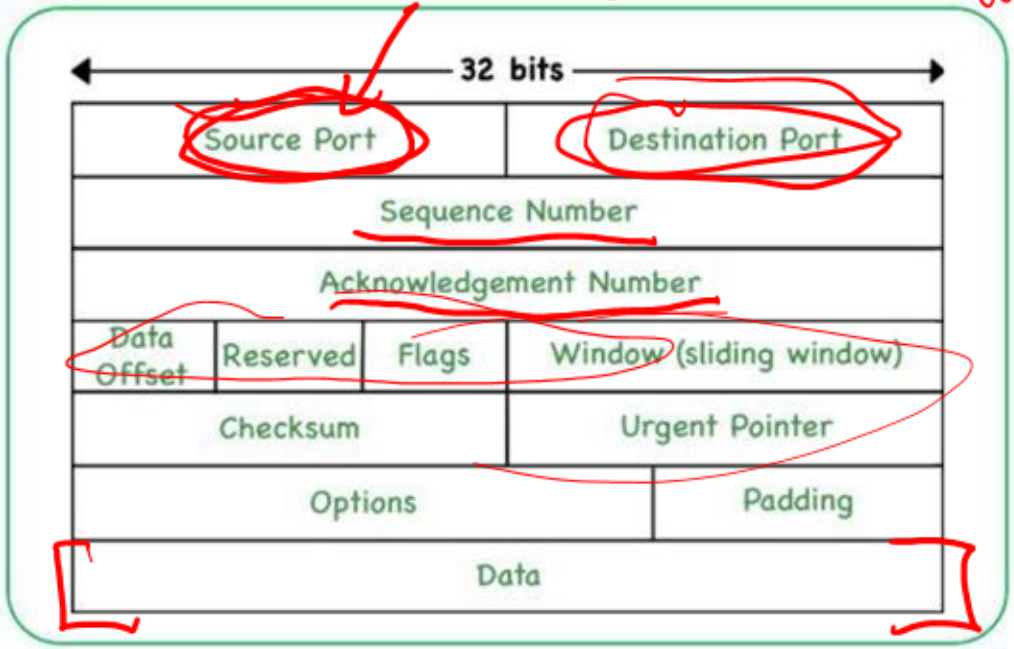
~~B) IP + TCP~~

~~C) Only TCP~~

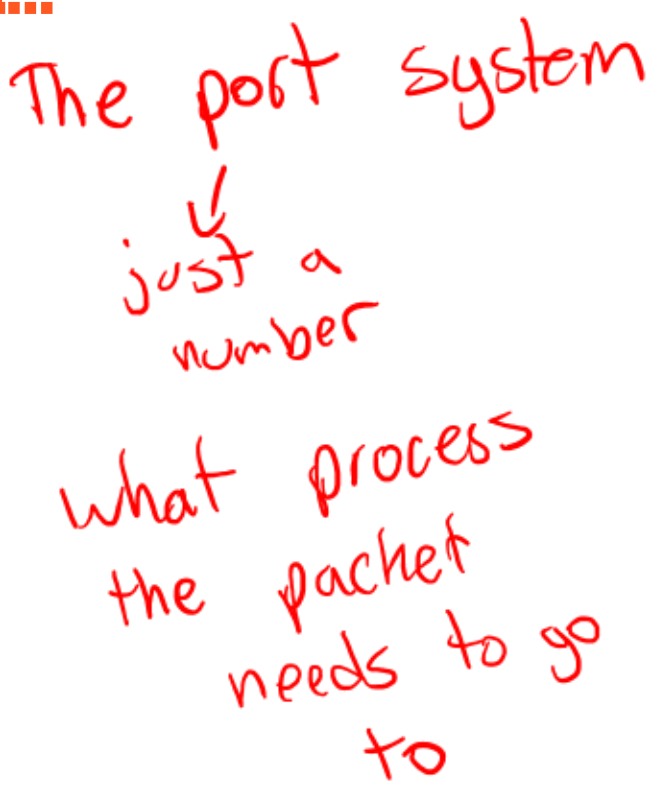
D) Only UDP

TCP Header Format

number of address on a computer linked to a process



The OS manages incoming information through...



Some ports are...

443 - your computer as a server listening for HTTPS request

ports 1024-49151 are free/open

Who manages the port system on your computer?

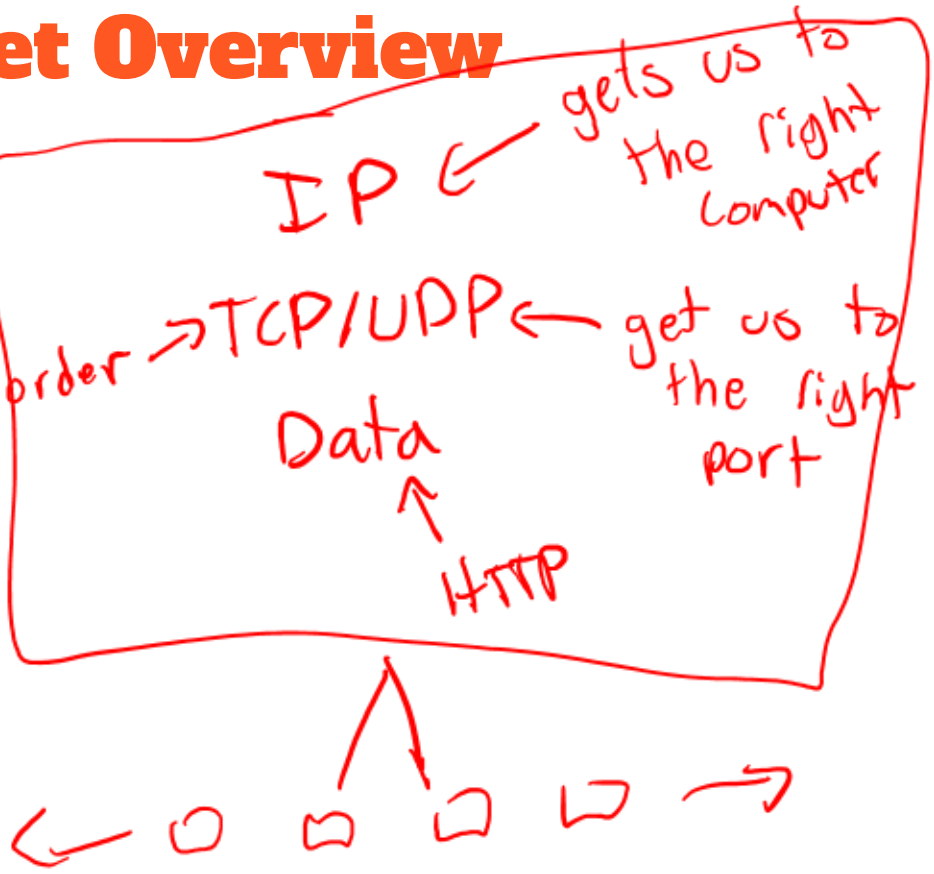
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A) The user
B) External servers
C) The OS
D) The internet

Packet Overview



Which are True?

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A) Packets are 1's and 0's

B) I need to use the IP protocol to send data over the internet.

C) I need to use the TCP protocol to send data over the internet

D) My OS uses ~~IP addresses~~ to figure out what process to send incoming packets to.

E) My LAN or home wifi has a private IP address only I know.

F) The internet is a system of computers

Client-Server Design

↑ rides ↓ rides → computer

server- a processs that listens for incoming requests
- waits for the client

Client - a process that initiates contact to ask for things from a server

Client-Server Design Example

